六人阿瓦隆中显式交流的作用

Silent Avalon Research Project Specification

CS7329 Reinforcement Learning 2026 10 20

1 项目概述

《》Silent Avalon

recurrent PPO “ ”
RTX 5070 Ti Laptop HPC 9B

“ ”

问题	需要回答的内 □	主 □ 证据
RQ1 Game Balance		
RQ2 Strategic Behavior		
RQ3 Implicit Signaling		

RQ1 RQ2 RQ3

2 文献定位与课程关联

DeepRole [1] “ ”
Silent Avalon

Is it Percival time yet SIGBOVIK 2017 66 5 61 [2]

STRATEGIST LLM PPO [3] O
“ ”

AvalonBench [4,10] Lan Avalon [5]

“ ” “ ”
RL

MDP Partially Observable Stochastic Game observation history
Actor π critic V return TD GAE advantage PPO [7,8] Self-play

RL [C1] Actor–Critic PPO [C2]

3 环境规则与信息边界

Merlin Loyal Servants Assassin Minion STRATEGIST Percival Morgana Servant Minion

项目	固定规范
	2 3 4 3 4
	Success Success Fail Fail
	Assassin Merlin
	Merlin
	Lady of the Lake

[9]

Communicative Avalon
“ j ”“ ”“ ” “ ”

Silent Avalon NULL

4 强化学习方法

recurrent Independent PPO MLP GRU actor–critic recurrent hidden state 128 128 GRU

action heads Assassin legal action mask 20

recurrent policy Markov state critic centralized critic actor

+1 −1 0 γ 1

→ → GAE advantage → PPO critic → checkpoint NULL actor padding

“ ”“ ” “ ” PPO GAE
pilot

Self play

checkpoint pilot 50% 25% 25%
on-policy

sanity baselines Merlin
IPPO [7]

5 实验矩阵与统计计划

实验	训练条 □	评估条件	解释
C → C			baseline
S → S			
C → S			
S → S mask			RQ3

S → S C → C C → S C → C checkpoint
S → S

2,000 pilot 20,000–100,000 B
5 10 3

PPO

C → C S → S C → S 1,000 15,000
5,000 5,000

95% bootstrap
Wilson

self-play

6 指标与可解释的结果

RQ1

ΔGood P(Good wins | S → S) − P(Good wins | C → C) 50%

P(Good wins) P( ) × [1 − P(Merlin |)]

Merlin 25%

RQ2

行为	建议量化方式	解释限制

行为	建议量化方式	解释限制
	Fail Fail	
	probe Merlin log loss	probe

checkpoint

7 隐式信号的证据标准

emergent communication [6]

probe “ ” “ ”

checkpoint GRU
placebo

“ ”

cross-play
Fail

probe RQ3 RQ1 RQ2

8 算力安排与自然语言扩展

HPC

RTX 5070 Ti Laptop recurrent PPO CPU GPU
GPU RAM

40–60 GPU pilot B 2 × × B ×
HPC HPC

AvalonBench engine [10]
Avalon DeepRole STRATEGIST

9B API

9B API NL-C
NL-S MARL

20 pilot token 100–300 64 token

9B FP16 18 GB 4-bit 4.5 GB KV cache
LLM RL

API token 50

“ RL” LLM RL

9 进度交付与失败处理

阶段	□ 议时间	验收内容
	10 20 –26	
	10 27 –11 2	Silent PPO pilot
	11 3 –16	C S checkpoint HPC
	11 17 –23	C → C S → S C → S RQ3
	11 24 –30	

mask credit assignment “ ”

C S RQ1 RQ2 RQ3

cross-play Percival Morgana

game id seed checkpoint

RQ3

self-play Nash equilibrium

□□□□

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[2] Nakamura, Y. (2017). Is it Percival time yet? A preliminary analysis of Avalon gameplay and strategy. SIGBOVIK 2017.

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[10] Light, J. et al. Avalon-LLM. AvalonBench STRATEGIST avalonbench_dev commit

[C1] CS7329 Fall 2026. Current Topics Presentation and Paper (Graduate Students). 2

[C2] Hahsler, M. Fall 2026 CS 7329 Reinforcement Learning Introduction_to_Reinforcement_Learning

DeepRole STRATEGIST O PPO recurrent actor–critic